The Green Revolution in India

Introduction to the Green Revolution

The Green Revolution refers to the dramatic increase in agricultural production, especially of wheat and rice, achieved through the use of high-yielding variety (HYV) seeds, chemical fertilisers, pesticides and expanded irrigation. It transformed India from a country dependent on food imports into a self-sufficient food producer. Globally, the term was first used in 1968 by William Gaud of the United States Agency for International Development.

Background: Why Did India Need It?

At independence, India's population of about 340 million faced chronic food shortages. The Bengal famine of 1943 killed about 2 to 3 million people. India was importing millions of tonnes of food grains, mainly wheat from the United States under the PL-480 programme, in what was called a "ship to mouth" existence. Severe droughts in 1965 and 1966, combined with wars with China in 1962 and Pakistan in 1965, brought India close to famine, and Prime Minister Lal Bahadur Shastri gave the slogan "Jai Jawan, Jai Kisan" and asked citizens to skip one meal a week.

When Did the Green Revolution Begin?

The Green Revolution in India began in the mid-1960s, usually dated to 1966 to 1967, when the government began to import and distribute high-yielding seeds of wheat on a large scale. It was launched under Agriculture Minister C. Subramaniam, who came from Tamil Nadu, and later Jagjivan Ram, with scientific leadership from M. S. Swaminathan, and was supported by Prime Ministers Lal Bahadur Shastri and Indira Gandhi. The first experiments in 1963 to 1965 used Mexican dwarf wheat varieties from Dr. Norman Borlaug. The 1967-68 wheat harvest, at about 17 million tonnes, was a record at the time, showing the success of the new approach. Wheat production increased from about 11 million tonnes in 1960 to about 36 million tonnes by 1980 and 55 million tonnes by 1990, and India's total foodgrain production grew from about 50 million tonnes in 1950-51 to more than 300 million tonnes in recent years. The Food Corporation of India was set up in 1965 and the Agricultural Prices Commission, now the Commission for Agricultural Costs and Prices, in 1965, to provide minimum support prices and procure grain.

Who Is Called the Father of the Green Revolution in India?

Dr. Mankombu Sambasivan Swaminathan, born in Kumbakonam, Tamil Nadu, on 7 August 1925, is called the Father of the Green Revolution in India. A geneticist and agricultural scientist, he worked with Dr. Norman Borlaug, who is called the Father of the Green Revolution worldwide and received the Nobel Peace Prize in 1970, to adapt high-yielding dwarf wheat varieties to Indian conditions, and he later led the Indian Agricultural Research Institute and the Indian Council of Agricultural Research. He won the first World Food Prize in 1987, which he used to establish the M. S. Swaminathan Research Foundation in Chennai in 1988. He chaired the National Commission on Farmers, whose recommendations in 2006 included setting minimum support prices at least 50 percent above the cost of production. He later advocated an "Evergreen Revolution", meaning sustainable increase in productivity without ecological harm. He died on 28 September 2023, at the age of 98, and was posthumously awarded the Bharat Ratna in 2024.

Key Elements

The main ingredients were high-yielding varieties of seeds, such as Kalyan Sona and Sonalika wheat and IR8 rice, nicknamed miracle rice, which was developed at the International Rice Research Institute in the Philippines and released in 1966; chemical fertilisers and pesticides; assured irrigation through canals, tube wells and pumps; farm mechanisation with tractors; credit and cooperative societies; minimum support prices; and agricultural research and extension through agricultural universities such as the one at Pantnagar and Tamil Nadu Agricultural University in Coimbatore.

Which Crops Benefited Most?

Wheat benefited the most in the first phase, with the biggest gains in Punjab, Haryana and western Uttar Pradesh, which are known as India's granary. Rice was the second main beneficiary, especially in Punjab, Andhra Pradesh and Tamil Nadu, where the delta districts of Thanjavur and the Cauvery basin adopted IR8 and the later ADT varieties. Other cereals such as maize, jowar and bajra also benefited somewhat, but pulses, oilseeds and millets saw little improvement, and in many places their area shrank as farmers shifted to wheat and rice. Cash crops such as cotton and sugarcane also saw higher yields in later years.

Other Revolutions

The Green Revolution inspired other movements: the White Revolution, or Operation Flood, launched in 1970 under Verghese Kurien, which made India the largest milk producer in the world; the Blue Revolution in fisheries; the Yellow Revolution for oilseeds; and the Golden Revolution in horticulture.

Problems and Criticism

The Green Revolution had serious side effects. Heavy use of fertilisers and pesticides polluted soil and water, and intensive irrigation lowered groundwater levels sharply, particularly in Punjab and Haryana. Continuous monoculture of rice and wheat reduced soil fertility and crop diversity and reduced the area of nutritious millets and pulses. The benefits went mostly to richer farmers in irrigated regions, increasing regional and economic inequality, and small farmers who could not afford inputs fell into debt. Greenhouse gas emissions from paddy fields and stubble burning increased, and reliance on a narrow set of crops raised food security and nutrition concerns.

Challenges and the Way Forward

Today India is a major exporter of rice and one of the largest producers of wheat. Experts recommend a "second Green Revolution" with sustainable practices such as drip irrigation, crop diversification towards millets and pulses, natural and organic farming, soil health cards, precision agriculture using drones and satellites, and climate-resilient crop varieties. The government's National Food Security Act of 2013 provides subsidised grain to about two-thirds of the population.

Quick Facts

- The Green Revolution in India began in the mid-1960s, around 1966 to 1967.
- M. S. Swaminathan, born in Kumbakonam, is called the Father of the Green Revolution in India; Norman Borlaug is the father globally.
- Wheat and rice benefited the most, especially in Punjab, Haryana and western Uttar Pradesh.
- IR8, the miracle rice, was released in 1966.
- Side effects include groundwater depletion, soil degradation and loss of crop diversity.